Basic Calculus

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ **Contents & learning path**

1. Calculus is the language of change and accumulation
2. Derivatives you should recognize
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1. Calculus is the language of change and accumulation

Astronomy is full of rates and totals. Velocity is the rate of change of position; luminosity is energy per unit time; flux is power per unit area; mass can be the integral of density over volume. Calculus turns “how fast?” and “how much in total?” into precise operations.

The derivative is a local slope:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}. \tag{1}$$

The definite integral is accumulated area:

$$\int_a^b f(x)\,dx = F(b) - F(a), \qquad F'(x) = f(x). \tag{2}$$

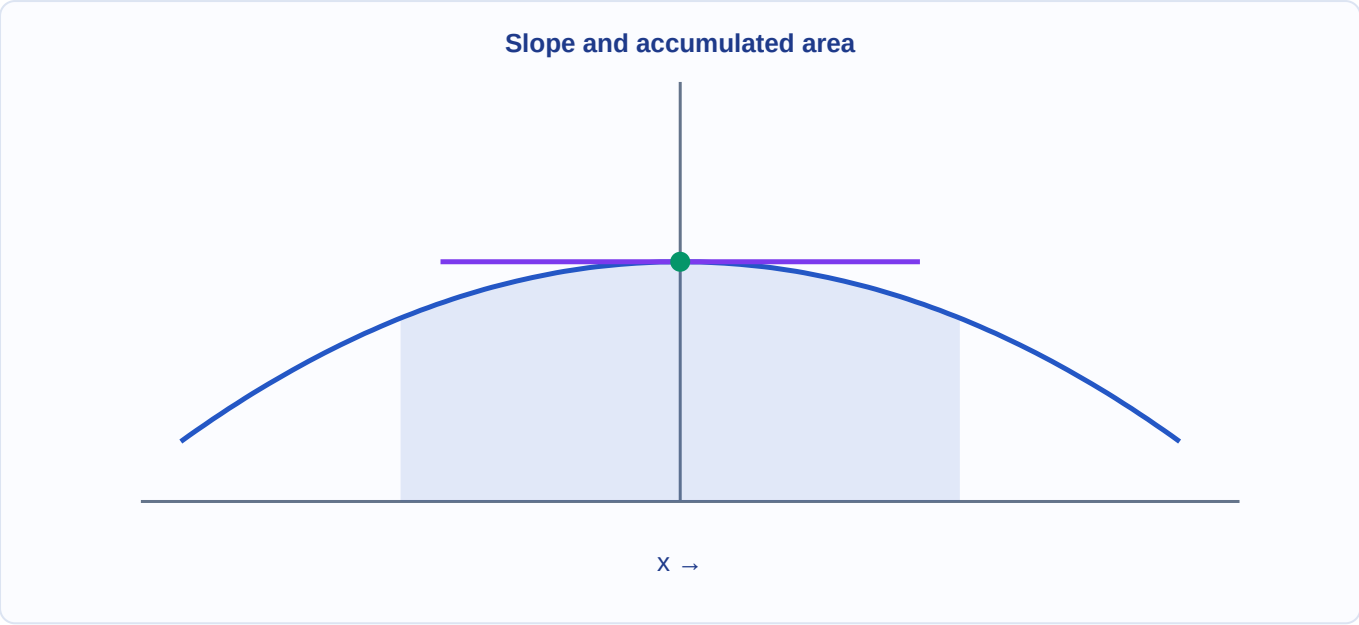


Figure 1. In astronomy, derivatives give rates and integrals give totals such as flux, mass or lookback time.

2. Derivatives you should recognize

Power rule: $d(x^n)/dx = nx^{n-1}$. Exponential: $d(e^{kx})/dx = ke^{kx}$. Trigonometric: $d(\sin x)/dx = \cos x$, $d(\cos x)/dx = -\sin x$. Product and chain rules let you differentiate combinations:

$$(uv)' = u'v + uv', \qquad \frac{d}{dx} f(g(x)) = f'(g(x))g'(x). \tag{3}$$

In physics, the chain rule is often the difference between a correct and an incorrect rate. If $r(t)$ changes, then $d(r^2)/dt = 2r\,dr/dt$, not merely $2r$.

3. Integrals you should recognize

Common antiderivatives include $\int x^n dx = x^{n+1}/(n+1) + C$ for $n \neq -1$, $\int e^{kx} dx = e^{kx}/k + C$, $\int \sin x\,dx = -\cos x + C$, and $\int dx/x = \ln|x| + C$. Definite integrals need limits and units. If f is a rate, the integral has units of rate times the integration variable.

Symmetry can save work. The integral of an odd function over a symmetric interval is zero. The average value of a function on $[a, b]$ is $(b-a)^{-1}\int_a^b f(x)\,dx$.

4. Small quantities and approximations

For small x in radians,

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}, \qquad (1+x)^n \approx 1 + nx. \tag{4}$$

These are Taylor approximations. They are not magic; they are the first terms of a local polynomial. The small-angle approximation is excellent for 1° (0.01745 rad), but poor for 30°.

5. Differential equations in astronomy

Many laws are differential equations. Radioactive decay obeys $dN/dt = -\lambda N$, giving $N = N_0 e^{-\lambda t}$. Newton's second law is $F = m\,d^2x/dt^2$. Hydrostatic equilibrium is $dP/dr = -\rho g$. The solution is a function, not a single number; initial or boundary conditions select the physical one.

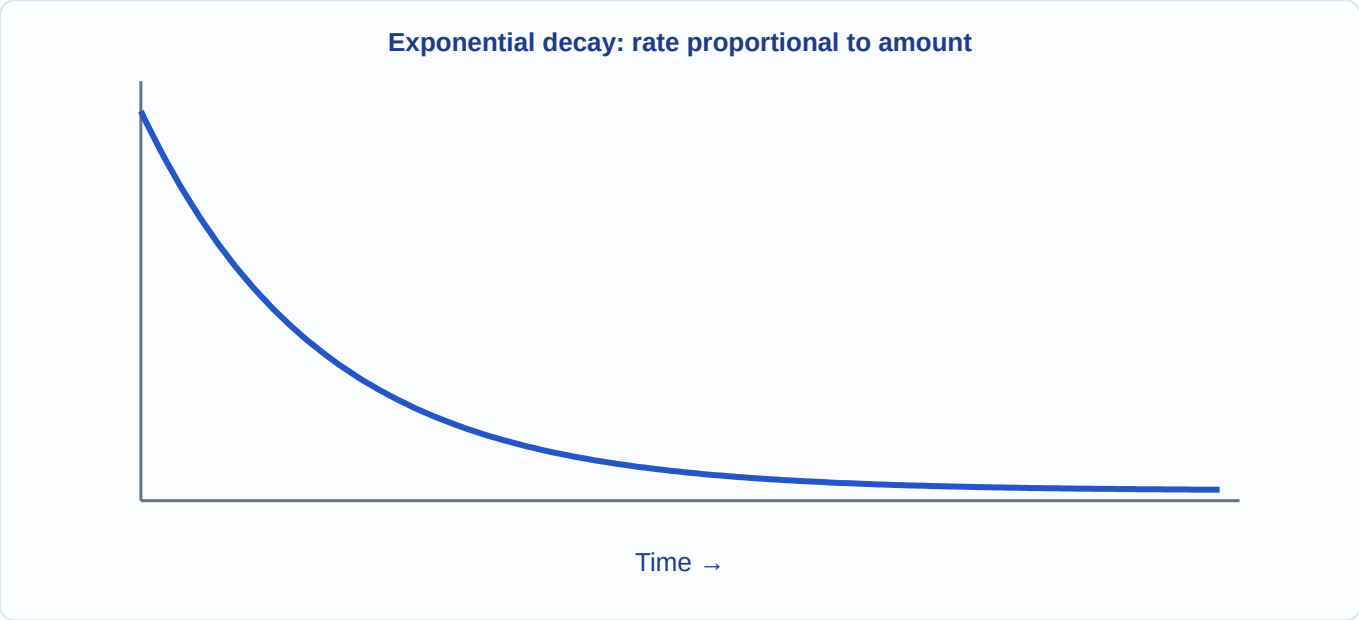


Figure 2. The half-life or e-folding time sets the scale; initial conditions set the normalization.

6. Optimization and physical reasoning

Maxima and minima often occur where a derivative is zero, but endpoints and physical constraints also matter. In orbital mechanics, the closest and farthest points are turning points of radius. In photometry, the minimum of a light curve may indicate an eclipse. Always check the second derivative or the physical context to decide whether a stationary point is a local maximum, local minimum or neither (for example, a point of inflection with horizontal tangent). "Saddle" is terminology for functions of more than one variable.

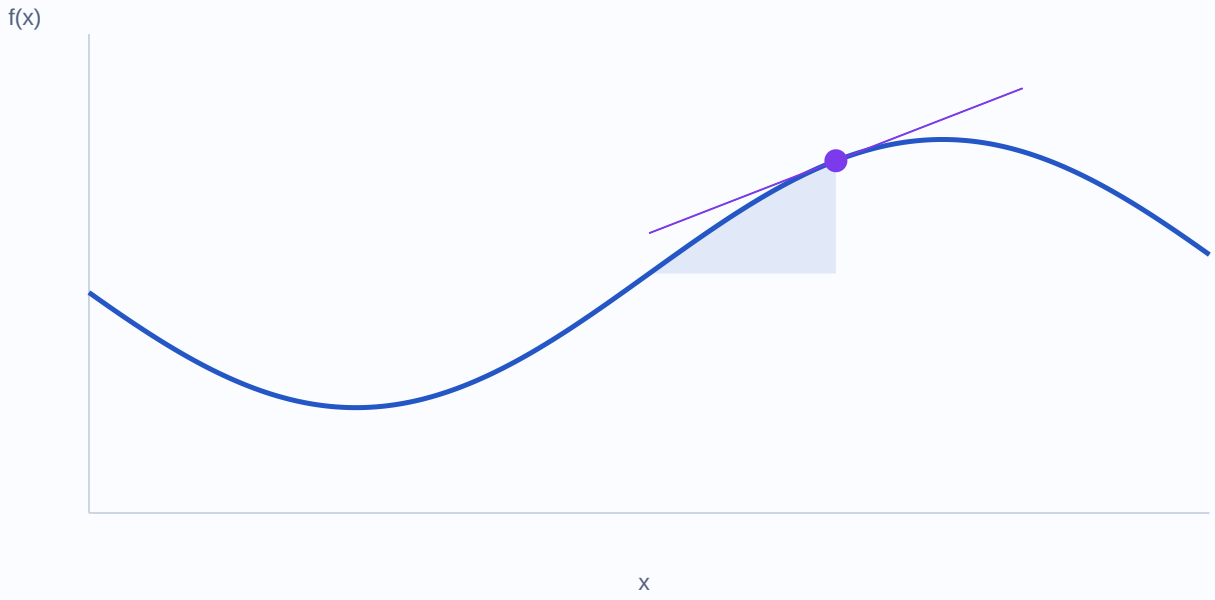
INTERACTIVE EXPERIMENT

A derivative is a local slope; an integral is accumulated area.

Compare a numerical derivative and integral with the exact result for the selected function. The shaded area is the integral from 0 to x.

f(x)	Numerical derivative	Exact derivative
0.8415	**0.5394**	**0.5403**

Numerical integral	Exact integral
0.4597	**0.4597**



The purple line is the tangent from the exact derivative. The shaded region is the signed integral from 0 to x.

Try this. Make h very large, then very small. When does the derivative become inaccurate, and why?

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Identify the variable, write the rate or accumulation law, differentiate or integrate with units, then check limiting cases and dimensions.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — A simple derivative. Differentiate $f(x) = 3x^4 - 2x + 7$. Evaluate the derivative at $x = 1$.

Worked solution →

[1] Problem 2 — A definite integral. Evaluate $\int_0^2 (3x^2 + 1)\,dx$.

Worked solution →

[2] Problem 3 — Chain rule. If $r(t) = 2t^2 + 1$, find $d(r^3)/dt$ at $t = 1$.

Worked solution →

[1] Problem 4 — Small angle. Use the small-angle approximation to estimate $\sin 2^\circ$. Compare with the exact value to three significant figures.

Worked solution →

[2] Problem 5 — Exponential decay. A sample decays with $\lambda = 0.1\ \text{yr}^{-1}$. Find the half-life and the fraction remaining after 10 years.

Worked solution →

[2] Problem 6 — Average value. Find the average value of $f(x) = x^2$ on $[0, 3]$.

Worked solution →

[2] Problem 7 — A maximum. A projectile's height is $h(t) = 20t - 5t^2$. Find the time and value of maximum height.

Worked solution →

[2] Problem 8 — Flux from luminosity. A star has luminosity $L(t) = L_0(1 + t/\tau)$. Find the total energy emitted from $t = 0$ to $t = \tau$.

Worked solution →

[3] Problem 9 — Taylor approximation. Use $(1 + x)^{1/2} \approx 1 + x/2$ to estimate $\sqrt{1.04}$. Estimate the size of the neglected term.

Worked solution →

[3] Problem 10 — A differential equation. Solve $dy/dt = ky$ with $y(0) = y_0$. If y doubles in 5 years, find k and $y(10)/y_0$.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.